

Housing Price Analysis Report

Exploratory Data Analysis of Residential Property Prices

Introduction

This project presents an Exploratory Data Analysis (EDA) of a housing dataset containing 545 residential properties and 13 variables. The objective was to examine the characteristics that influence house prices, identify trends and relationships within the data, and provide insights that can support informed home-buying decisions. Before the analysis, the dataset was inspected for missing values, duplicate records, outliers, and inconsistent categorical values to ensure the quality and reliability of the results.

Key Findings

The analysis produced several important insights into the housing market:

- **House size has a positive relationship with price.** The correlation analysis and scatter plot show that larger houses generally sell for higher prices, making area one of the strongest predictors of house value.
- **Location significantly influences house prices.** Houses located in preferred areas have a higher average selling price than those outside preferred areas, indicating that buyers are willing to pay a premium for desirable locations.
- **Property features add value.** Houses with air conditioning and those that include both a basement and air conditioning have higher average prices than houses without these features. This suggests that additional amenities contribute to increased market value.
- **Three-bedroom houses are the most common.** More than half of the houses in the dataset have three bedrooms, indicating that this property type dominates the housing market and may represent the typical family home.

Visualization 1 – Area vs. House Price Scatter Plot

[Insert Scatter Plot screenshot here]

Figure 1. The scatter plot illustrates a positive relationship between house area and selling price. Although some variation exists, larger houses generally command higher prices, showing that property size is a major factor influencing market value.

Visualization 2 – Correlation Heatmap

[Insert Correlation Heatmap screenshot here]

Figure 2. The correlation heatmap summarizes the relationships between the numerical variables in the dataset. It confirms that area has one of the strongest positive correlations with house price, while variables such as bedrooms, bathrooms, and stories also show positive relationships with property value.

Conclusion and Recommendations

The exploratory data analysis identified several important factors that influence housing prices, including property size, location, and additional amenities. The dataset was found to be clean and suitable for analysis, with no missing values or duplicate records. Although statistical outliers were identified in the price and area variables, they were retained because they likely represent genuine high-value properties rather than data errors.

Based on the findings, the following recommendations are made for a hypothetical house buyer seeking value:

- Prioritize houses with larger living areas, as house size has one of the strongest positive relationships with property price and long-term value.
- Compare properties inside and outside preferred areas. Homes outside preferred areas may offer similar features at a lower price, providing better value for budget-conscious buyers.

- Consider additional features such as air conditioning and basements only when they match your needs, as these amenities increase property value but also increase purchase cost.

Overall, this analysis demonstrates how exploratory data analysis can transform raw housing data into meaningful insights that support better purchasing decisions and provide valuable information for buyers, sellers, and real estate professionals.